

Chapter-01

Network topology refers to how devices in a network are physically or logically arranged. Here's an overview of the main types—**Star, Bus, Ring, Mesh, and Hybrid**—including their concepts, advantages, and disadvantages.

1. ★ Star Topology

- **Concept:** All devices are connected to a central hub, switch, or router. The central device acts as a mediator for data transmission.

✓ Advantages:

- **Easy to install and manage.**
- **Failure isolation:** If one node fails, it doesn't affect the rest of the network.
- **Scalable:** Easy to add new devices.
- **Efficient performance** due to direct connections.

✗ Disadvantages:

- **Single point of failure:** If the central hub fails, the entire network goes down.
 - **Higher cost** due to the central device and more cabling.
 - **Limited distance** due to central dependency.
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2. 🚌 Bus Topology

- **Concept:** All devices share a single backbone cable. Data is sent in both directions, and each device checks if the data is intended for it.

✓ Advantages:

- **Cost-effective** due to less cabling.
- **Simple to set up** for small networks.
- **Requires less cable** compared to star topology.

✗ Disadvantages:

- **Difficult to troubleshoot** when problems occur.
 - **Performance issues:** Network slows down with heavy traffic.
 - **Not scalable:** Adding devices can cause disruption.
 - **Single point of failure:** If the main cable breaks, the whole network fails.
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3. Ring Topology

- **Concept:** Devices are connected in a circular fashion. Each device has exactly two neighbors. Data travels in one direction (or both in dual-ring).

✓ Advantages:

- **Data flows in one direction**, reducing collisions.
- **Good for handling high-volume traffic.**
- **Easier fault isolation** than bus topology.

✗ Disadvantages:

- **Failure in one device** can affect the entire network.
 - **Difficult to reconfigure** or expand.
 - **Slower** compared to star and mesh due to sequential data travel.
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4. Mesh Topology

- **Concept:** Every device is connected to every other device. It can be **full mesh** (all devices connected) or **partial mesh** (only some devices connected).

✓ Advantages:

- **Highly reliable and fault-tolerant**—if one link fails, data can take another route.
- **High security** due to dedicated point-to-point connections.
- **Scalable** for high-performance networks.

✗ Disadvantages:

- **Complex and expensive** due to extensive cabling.
 - **Challenging to set up** and maintain.
 - **Not suitable for small networks.**
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5. Hybrid Topology

- **Concept:** A combination of two or more different topologies (e.g., star-bus, star-ring). It leverages the strengths of each.

✓ Advantages:

- **Highly flexible and scalable.**
- **Efficient for large organizations.**
- **Fault-tolerant** depending on design.

✖ Disadvantages:

- **Complex to design and manage.**
 - **Higher cost** due to multiple topology integration.
 - **Requires advanced management** to handle.
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✔ Summary Comparison:

Topology	Cost	Reliability	Scalability	Ease of Setup
Star	Medium	Moderate	Easy	Easy
Bus	Low	Low	Difficult	Easy
Ring	Medium	Moderate	Difficult	Moderate
Mesh	High	High	Moderate	Complex
Hybrid	High	High	High	Complex

Each topology serves different needs—**Star** is common in homes and small offices, **Mesh** is used in critical networks, and **Hybrid** fits complex organizational requirements.

Definition of Data Communication Systems and Major Components (4 Marks)

Data Communication System:

A **data communication system** is a system that enables the exchange of data between two or more devices through a transmission medium such as cables, wireless signals, or fiber optics. The main objective is to transfer data accurately and efficiently.

Major Components of a Data Communication System:

1. **Message:**
 - The actual data or information that needs to be communicated (e.g., text, audio, video).
2. **Sender:**
 - The device that initiates and sends the data (e.g., computers, phones).
3. **Receiver:**
 - The device that receives and processes the transmitted data.
4. **Transmission Medium:**
 - The physical path through which the data is transmitted (e.g., copper wires, fiber optics, or wireless channels).

5. Protocol:

- A set of rules and standards that govern data communication, ensuring that devices can understand and interpret the data correctly.

Protocols and Standards in Data Communication & Key Elements of a Protocol (5 Marks)

Protocols:

A **protocol** is a set of rules and conventions that govern how data is transmitted and received over a network. Protocols ensure proper communication between devices by defining data formats, error handling, and data compression methods.

Standards:

Standards are established guidelines and specifications that ensure consistency and compatibility among different devices and technologies in data communication. For example, **IEEE, ISO, and ITU** are organizations that set communication standards.

Key Elements of a Protocol:

1. Syntax:

- Refers to the **structure and format** of the data, including the arrangement of bits and bytes.
- For example, how data packets are formed.

2. Semantics:

- Defines the **meaning of each section** of bits and the purpose of each message.
- It helps in understanding what action to take for each message.

3. Timing:

- Deals with the **synchronization and speed** of data transmission.
- It ensures that both sender and receiver are synchronized and can handle data flow correctly.

Chapter-02

OSI model layers and their functions:

1. Physical Layer

The physical layer is responsible for movements of individual bits from one hop (node) to the next

- Representation of bits
- Synchronization of bits
- Line configuration
- Data rate
- Physical topology
- Transmission mode

2. Data Link Layer

The data link layer is responsible for moving frames from one hop (node) to the next.

- Framing
- Physical addressing
- Flow control
- Access control
- Error control

3. Network Layer

The network layer is responsible for the delivery of individual packets from the source host to the destination host.

- Logical addressing
- Routing

4. Transport Layer

The transport layer is responsible for the delivery of a message from one process to another.

- Service-point addressing
- Segmentation and reassembly
- Connection control
- Flow control
- Error control

5. Session Layer

The session layer is responsible for dialog control and synchronization.

- Dialog control.
- synchronization.

6. Presentation Layer

The presentation layer is responsible for translation, compression, and encryption.

- Translation
- Compression
- Encryption

7. Application Layer

The application layer is responsible for providing services to the user.

- Network virtual terminal
- File transfer, access, and management.
- Mail services.
- Directory services.

Addresses in TCP/IP

Four levels of addresses are used in an internet employing the TCP/IP protocols: physical, logical, port, and specific.

Physical Address

A physical address refers to a unique identifier assigned to a device on a network, typically in the form of a MAC (Media Access Control) address. This address is used at the Data Link layer of the OSI model and allows devices to communicate within a local network.

Format: A MAC address is typically represented as a 48-bit (6-byte) physical address, consisting of 12 hexadecimal digits grouped into six pairs (e.g., 00:1A:2B:3C:4D:5E), with each byte (2 digits) separated by a colon

Logical Address

A **logical address** is a unique identifier assigned to a device at the **Network Layer** of the OSI model, allowing communication across different networks. Unlike physical addresses, logical addresses are not hardcoded into hardware and can be changed.

- **Function:** Used for identifying and locating devices on a network, enabling communication across different networks.
- **Example:** IP addresses (IPv4 or IPv6) are the most common form of logical addresses.
- **Difference from Physical Address:** While a MAC address is fixed for a device, an IP address can be reassigned dynamically (e.g., using DHCP).

Port Addresses

A port address is a number that identifies specific services or applications on a device connected to a network. It works with an IP address to ensure data reaches the correct application.

A 16-bit port address represented as one single number.Ex:753.

Function: Port addresses route data to the correct application, enabling effective communication. In short, port addresses are crucial for directing network traffic to the right services on devices.

Specific Addressing

Specific addressing refers to the process of directing data to a particular device or application within a network. It ensures that the correct recipient gets the intended data.

- **Types of Specific Addressing:**
 1. **Unicast Addressing:** Data is sent to a **single specific** destination device.
 - Example: Sending an email to a single recipient.
 2. **Broadcast Addressing:** Data is sent to **all devices** in a network.
 - Example: ARP request to find a MAC address.
 3. **Multicast Addressing:** Data is sent to **multiple devices** that are part of a specific group.
 - Example: Streaming a live video to selected users.
 4. **Anycast Addressing:** Data is sent to **the nearest device** in a group.
 - Example: Content delivery networks (CDNs) directing users to the closest server.

Chapter -03

Transmission Impairment

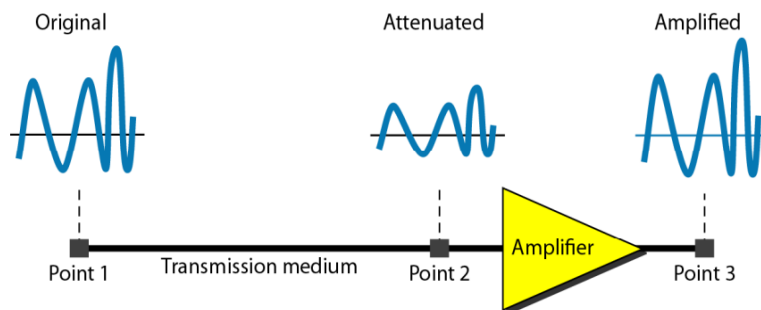
When a signal travels through a medium (like a wire, fiber optic cable, or wireless channel), it does not always remain perfect. Various factors cause impairments that degrade the quality of the signal, leading to errors or loss of information.

Types of Transmission Impairments:

1. Attenuation (Signal Loss):

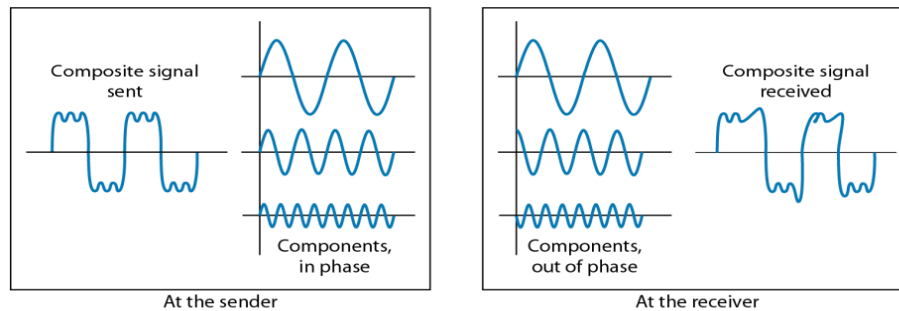
- **Definition:** Attenuation is the weakening of a signal as it travels through a medium, like a wire or the air. The further the signal travels, the weaker it gets.

- **Cause:**
 - **Resistance in cables:** When signals travel through wires or cables, the resistance causes energy loss.
 - **Long distances:** The longer the distance, the more the signal weakens.
 - **Atmospheric absorption:** In wireless communication, signals can get absorbed or weakened by the atmosphere, especially over long distances.
- **Solution:**
 - **Amplifiers:** These devices are used to boost analog signals back to their original strength.
 - **Repeaters:** In digital communication, repeaters receive the signal, clean it, and then send it forward, restoring signal quality over long distances.



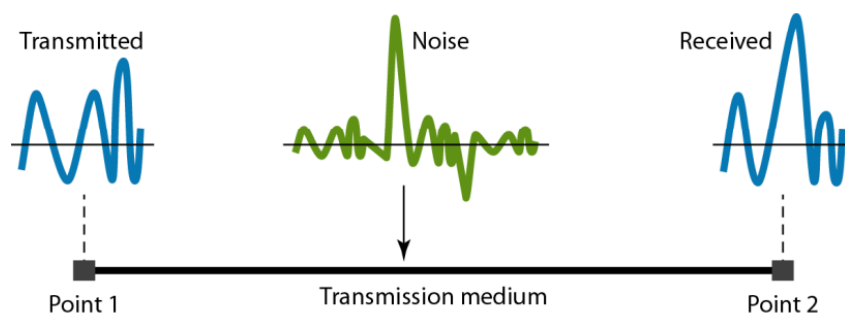
2. Distortion:

- **Definition:** Distortion occurs when the shape of the signal changes, usually because different parts of the signal travel at different speeds. It happens most often when transmitting over long distances or through certain materials.
- **Cause:**
 - **Frequency-dependent propagation speed:** Different frequencies in the signal travel at different speeds in the transmission medium (like cables or air), causing the signal to change shape over time.
 - For example, high-frequency signals might travel faster than low-frequency signals, causing the signal to lose its original shape.
- **Solution:**
 - **Equalizers:** These devices are used to adjust the signal and reduce the difference in delay for different frequencies. This helps to restore the original signal shape, making it easier to interpret.



3. Noise:

- **Definition:** Noise is unwanted random signals that interfere with the original signal, making it harder to detect the real data.
- **Cause:**
 - **External sources:** Electrical equipment, motors, or even weather conditions can create noise that affects communication channels.
 - **Internal sources:** Even the cables and devices used to transmit signals can create noise, especially if they are not properly shielded.
- **Solution:**
 - **Shielding:** Using properly insulated cables can help reduce the interference.
 - **Error-correcting codes:** These can help detect and correct noise-related errors in digital data transmission.
 - **Filtering:** Noise filters can help reduce unwanted signals in both analog and digital communications.



What does the Nyquist theorem have to do with communications?

The **Nyquist Theorem** in communications helps determine the **maximum bit rate** of a channel based on its **bandwidth** and the **number of signal levels**. It states that the bit rate can be calculated using the formula:

$$R = 2 \times B \times \log_2(L)$$

Where:

- R is the bit rate in bps,
- B is the bandwidth in Hz,
- L is the number of signal levels.

The theorem is significant because it helps determine how efficiently a communication channel can transmit data. A higher number of signal levels increases the potential bit rate, allowing more data to be transmitted within the same bandwidth, although this requires more complex encoding and decoding systems. This is crucial for designing communication systems that can handle high-speed data transmission.

Chapter-04

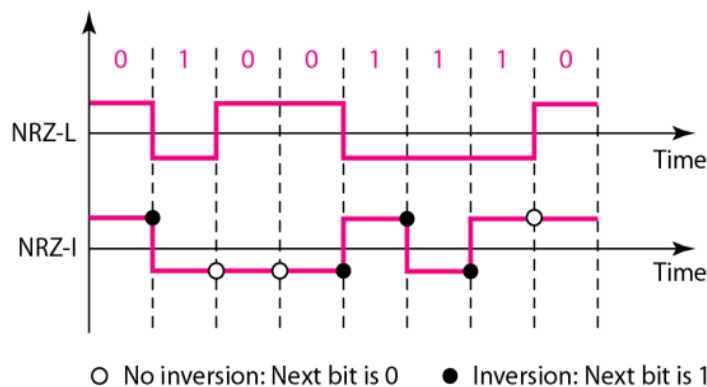
Polar NRZ-L and Polar NRZ-I are two types of Non-Return-to-Zero (NRZ) line coding schemes used in digital communication. Both belong to the polar encoding method, meaning they use both positive and negative voltage levels to represent binary data instead of staying only positive (unipolar) or using three levels (bipolar).

1. NRZ-L (Non-Return-to-Zero Level)

- **Encoding Rule:**
 - The signal level depends directly on the bit value.
 - 0 → High level (positive voltage)
 - 1 → Low level (negative voltage)
- **Behavior in Figure:**
 - At 0, the signal remains high.
 - At 1, the signal goes low.
 - This behavior continues throughout the sequence.

2. NRZ-I (Non-Return-to-Zero Inverted)

- **Encoding Rule:**
 - A 1 causes an inversion (change) in the signal.
 - A 0 means the signal stays the same.
- **Behavior in Figure:**
 - At 1, the signal flips (black dot indicates inversion).
 - At 0, the signal remains unchanged (white dot indicates no inversion).



Comparison Between NRZ-L and NRZ-I

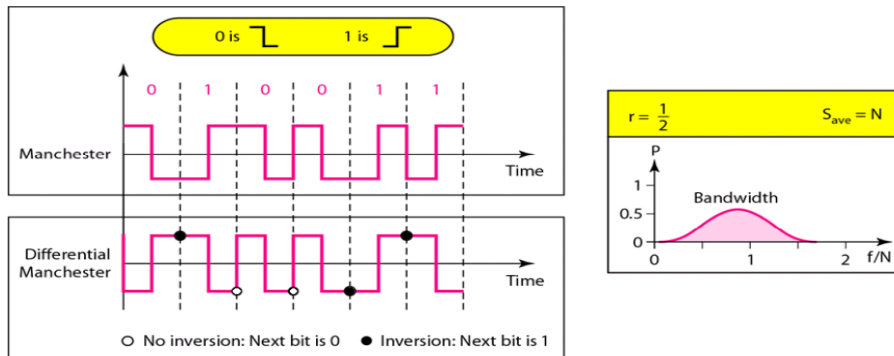
Feature	NRZ-L (Non-Return-to-Zero Level)	NRZ-I (Non-Return-to-Zero Inverted)
Encoding Rule	Voltage level represents bit value (0 = High, 1 = Low).	Bit 1 causes inversion; bit 0 keeps the same level.
Signal Transition	Changes based on bit values.	Changes only when a 1 is encountered.
Synchronization	More difficult to synchronize due to long sequences of 0s or 1s.	Easier to synchronize due to transitions at 1s.
Noise Immunity	Less immune to noise because no transition may occur for a long time.	More immune to noise due to frequent transitions.
Power Efficiency	Less efficient because voltage changes often.	More efficient as transitions occur only for 1s.
Usage	Used in older systems, e.g., early Ethernet.	Used in modern data transmission like USB, SD cards.
Bandwidth Requirement	Higher, as signal changes frequently.	Lower, as changes occur only for 1s.

1. Manchester Encoding:

- Uses **self-clocking**, meaning a transition occurs in every bit period, ensuring synchronization.
- Each bit has **two signal levels**:
 - "0" → Transition from high to low (↓)
 - "1" → Transition from low to high (↑)
- Provides good synchronization but requires double the bandwidth compared to NRZ schemes.
- Used in **Ethernet (IEEE 802.3)**.

2. Differential Manchester Encoding:

- A variation of Manchester encoding that depends on **changes between bits rather than absolute levels**.
- **Rules**:
 - Always a transition at the middle of each bit period (clocking transition).
 - "0" → An additional transition at the start of the bit period.
 - "1" → No additional transition at the start.
- More reliable in **noisy environments** since decoding depends on changes, not levels.
- Used in **Token Ring networks (IEEE 802.5)**.



5. An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements are needed?

- Bit rate (R_b) = 8000 bps
- Baud rate (S) = 1000 baud
- Data elements per signal element:

$$r = \frac{R_b}{S} = \frac{8000}{1000} = 8$$

Answer: Each signal element carries **8 bits**.

- Number of signal elements needed:

$$S = 1000$$

Since each symbol carries **8 bits**, we need **1000 symbols per second** to represent 8000 bits.

1. Define line coding. Apply Manchester and Differential Manchester schemes on 10110100 to generate digital signals.

Line coding is the process of converting digital data into a digital signal for transmission. It determines how binary 1s and 0s are represented in the signal.

Manchester Encoding:

- Each bit has a transition in the middle of the bit period.
- '1' → Low to High, '0' → High to Low (IEEE 802.3 convention).
- For 10110100:

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1 → Low to High
0 → High to Low
1 → Low to High
1 → Low to High
0 → High to Low
1 → Low to High
0 → High to Low
0 → High to Low
```

Differential Manchester Encoding:

- Always a transition at the start of the bit period for synchronization.
- '1' → No transition in the middle, '0' → Transition in the middle.
- For 10110100:

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1 → No transition
0 → Transition
1 → No transition
1 → No transition
0 → Transition
1 → No transition
0 → Transition
0 → Transition
```

2. A signal carries data where one data element is encoded as two signal elements. If the bit rate is 10 kbps, what is the average baud rate if the case factor is between 0 and 1?

Solution

We assume that the average value of c is $1/2$. The baud rate is then

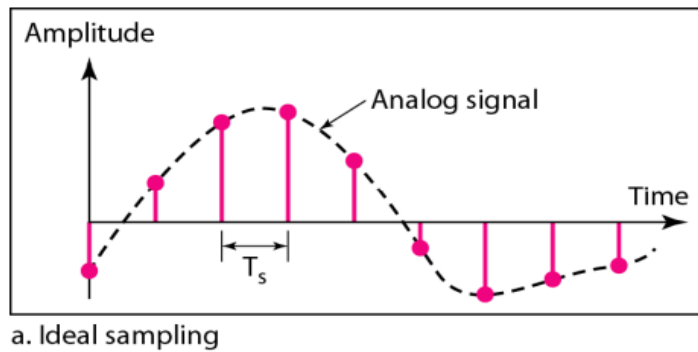
$$S = c \times N \times \frac{1}{r} = \frac{1}{2} \times 100,000 \times \frac{1}{1} = 50,000 = 50 \text{ kbaud}$$

Describe different types of sampling method use in pulse code modulation

Sampling is the process of converting a continuous signal into a discrete signal by measuring its amplitude at regular time intervals.

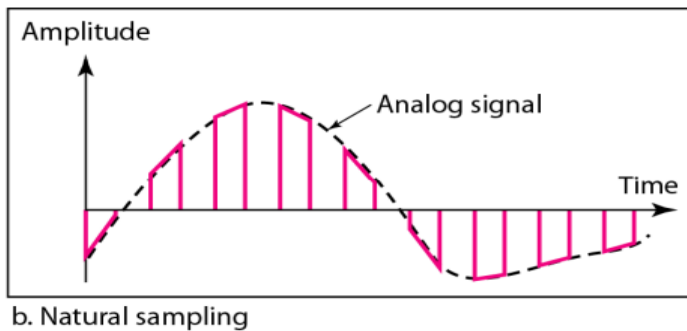
1. Ideal Sampling (Impulse Sampling)

- Also called **instantaneous sampling**.
- The analog signal is sampled using **impulses of zero width and infinite height** at regular intervals.
- It's a theoretical concept because ideal impulses don't exist in the real world.
- **Key Point:** Produces perfect samples but is impractical for real systems.



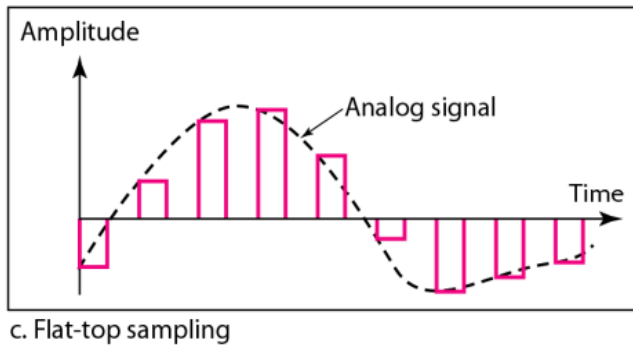
2. Natural Sampling

- Samples are taken by multiplying the signal with a **train of rectangular pulses**.
- The sampled value follows the **actual shape of the signal** during the sampling interval.
- More practical than ideal sampling but still complex in implementation.



3. Flat-Top Sampling (Practical Sampling)

- A refined version of natural sampling.
- The sampled signal is held at a **constant value (flat-top)** for a short duration.
- It simplifies processing, as the sample value doesn't vary within the sampling interval.
- **Key Point:** This method is widely used in PCM systems as it's easier for the analog-to-digital converter (ADC) to process.



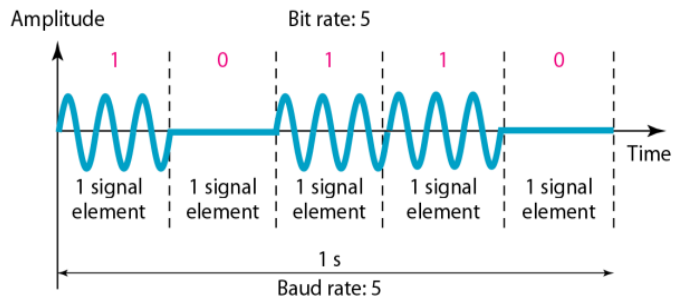
4. Compare Pulse Code Modulation (PCM) and Delta Modulation (DM).

Feature	PCM (Pulse Code Modulation)	DM (Delta Modulation)
Bit Representation	Uses multiple bits per sample	Uses 1 bit per sample
Quantization Levels	Multiple levels	Two levels ($\pm\Delta$)
Bandwidth Usage	Higher due to multi-bit representation	Lower due to 1-bit encoding
Complexity	More complex	Simpler
Error Handling	More resistant to noise	More prone to slope overload & granular noise
Application	High-quality digital audio & telephony	Low-bandwidth voice transmission

Processes of ASK, FSK, and PSK

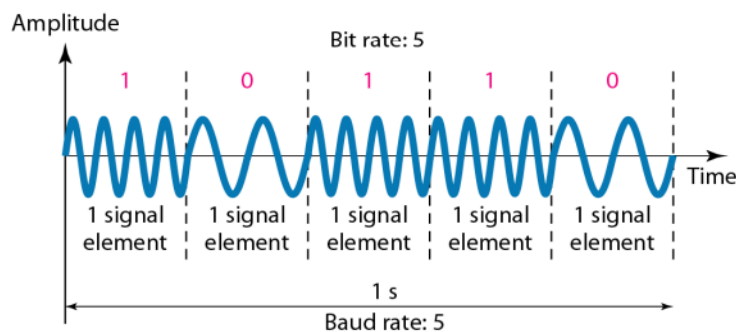
1. Amplitude Shift Keying (ASK) Process

1. **Input Binary Data:** A digital signal (1s and 0s) is given as input.
2. **Carrier Signal Generation:** A high-frequency sine wave is used as the carrier.
3. **Modulation:**
 - When input = **1**, the carrier is transmitted at full amplitude.
 - When input = **0**, the carrier is **absent or at lower amplitude**.
4. **Transmitted Signal:** The amplitude of the carrier signal varies according to the input data.



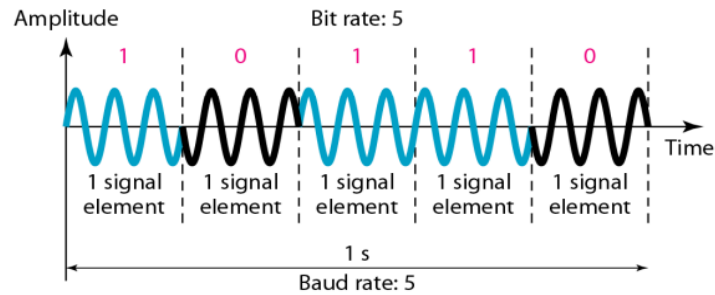
2. Frequency Shift Keying (FSK) Process

1. **Input Binary Data:** A digital signal with 1s and 0s is used.
2. **Carrier Signal Generation:** Two different frequencies are chosen (f_1 for '1' and f_2 for '0').
3. **Modulation:**
 - When input = **1**, a higher frequency (f_1) is transmitted.
 - When input = **0**, a lower frequency (f_2) is transmitted.
4. **Transmitted Signal:** The frequency of the carrier changes based on input bits.



3. Phase Shift Keying (PSK) Process

1. **Input Binary Data:** Digital input signal (1s and 0s).
2. **Carrier Signal Generation:** A constant-frequency sine wave is generated.
3. **Modulation:**
 - The **phase** of the carrier signal is shifted to represent 1s and 0s.
 - Example: In **BPSK**, a 180° phase shift is used between 1 and 0.
 - In **QPSK**, two bits are mapped to four different phase angles.
4. **Transmitted Signal:** A phase-modulated wave is sent for transmission.



Binary Amplitude Shift Keying (BASK) and Its Implementation

What is BASK?

Binary Amplitude Shift Keying (BASK) is a digital modulation technique in which the amplitude of a carrier signal is varied according to the binary data (1s and 0s) being transmitted. It is one of the simplest forms of Amplitude Shift Keying (ASK).

How BASK Works?

- A high-frequency carrier signal is used for transmission.
- When the input binary data is '1', the carrier signal is transmitted at full amplitude.
- When the input binary data is '0', the carrier signal is completely turned off (zero amplitude).
- This results in a signal that appears as bursts of the carrier wave for 1s and no signal for 0s.

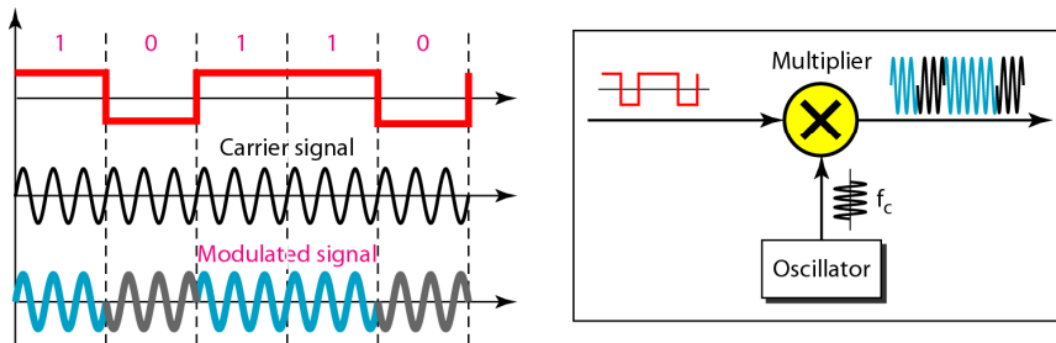
Implementation of BASK (Refer to Diagram)

1. **Input Binary Data:** The data to be transmitted is a sequence of 1s and 0s.
2. **Carrier Signal:** A continuous high-frequency sinusoidal wave is used as a carrier.

3.Modulation Process:

1. The binary data (red signal in the diagram) is fed into a multiplier circuit.
2. A local oscillator generates the carrier signal.
3. The multiplier controls the amplitude of the carrier: it **allows the carrier through for 1s** and **blocks it for 0s**.

4.Resulting Modulated Signal: The output is a modulated waveform where the carrier appears for binary '1' and disappears for binary '0'



Quadrature Phase Shift Keying (QPSK) and Its Implementation

What is QPSK?

Quadrature Phase Shift Keying (QPSK) is a digital modulation technique in which data is transmitted by changing the **phase** of a carrier signal. Unlike Binary Phase Shift Keying (BPSK), which transmits 1 bit per symbol, QPSK transmits **2 bits per symbol**, making it **twice as efficient**.

Process of QPSK (Quadrature Phase Shift Keying)

1. Input Binary Data:

- The incoming data is in binary form (0s and 1s).
- QPSK processes **two bits per symbol** to improve efficiency.

2. Bit Pairing & Mapping:

- The binary input is grouped into pairs (e.g., **00, 01, 10, 11**).
- Each pair is mapped to a unique **phase shift** (e.g., -135° , -45° , 135° , 45°).

3. Carrier Signal Generation:

- A high-frequency **carrier wave** is generated.
- Two carrier signals are used:
 - **In-phase (I) carrier** → Original carrier signal.
 - **Quadrature (Q) carrier** → Same signal but phase-shifted by 90° .

4. Multiplication (Modulation):

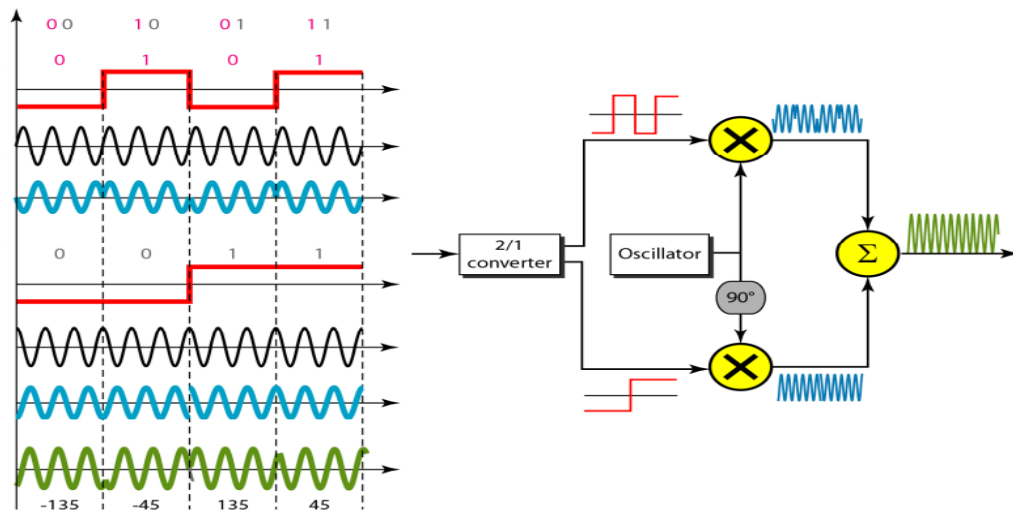
- The I and Q components are multiplied with their respective carrier signals.
- The I-component modulates the **cosine** wave.
- The Q-component modulates the **sine** wave (shifted by 90°).

5. Summation:

- The two modulated signals (I and Q) are summed to produce the **final QPSK-modulated signal**.

6. Transmission:

- The combined QPSK signal is sent over the communication channel.
- The receiver demodulates it by separating I and Q components and mapping them back to binary data.



7. A telephone subscriber line must have an SNR_{dB} above 40. What is the minimum number of bits required to represent or transmit each sample?

Using Shannon's formula for the number of quantization levels:

$$N = \log_2(1 + SNR)$$

Given $SNR_{dB} = 40$, we convert it to linear form:

$$SNR = 10^{(40/10)} = 10^4 = 10000$$

$$N = \log_2(1 + 10000) = \log_2(10001)$$

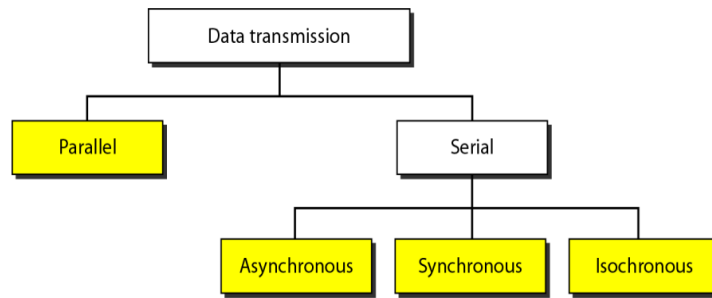
Approximating,

$$N \approx \log_2(2^{13.3}) \approx 13.3$$

Since N must be an integer, we take $N = 14$ bits.

Answer: 14 bits per sample.

Data Transmission is the process of sending digital or analog data from one device to another over a communication medium, such as cables, optical fibers, or wireless signals. It enables information exchange between devices, networks, or systems



Chapter-07

Transmission Media

Transmission media refer to the physical pathways that connect computers, devices, and people on a network. These media are used to transmit data from one location to another and can be categorized into two main types: **Guided Media** and **Unguided Media**.

1. Guided Media (Wired Media)

Guided media use physical conductors to transmit data signals. The signals are confined to the medium, and their path is defined.

Types of Guided Media:

1. Twisted-Pair Cable:

- **Description:** Consists of two insulated copper wires twisted together to reduce electromagnetic interference.
- **Types:**
 - **Unshielded Twisted Pair (UTP):** Commonly used in Ethernet networks.
 - **Shielded Twisted Pair (STP):** Includes shielding to reduce interference further.
- **Example:** Ethernet cables like Cat5, Cat6.
- **Performance:**

- Moderate data transmission speed (up to 10 Gbps for Cat6a).
 - Limited distance (up to 100 meters for Ethernet).
 - **Applications:**
 - LANs, office networks, and telephone lines.
 - Used in networking for homes, offices, and data centers.
-

2. Coaxial Cable:

- **Description:** Consists of a central core conductor, insulating layer, metallic shield, and outer cover.
 - **Example:** RG-6 cable for cable TV.
 - **Performance:**
 - Higher bandwidth than twisted-pair cables.
 - More resistant to EMI due to the shielding.
 - Can transmit data over longer distances compared to UTP.
 - **Applications:**
 - Used for cable television, broadband internet, and traditional telephone networks.
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3. Fiber-Optic Cable:

- **Description:** Uses light signals to transmit data at high speeds with minimal signal loss.
- **Types:**
 - **Single-mode Fiber:** For long-distance communication.
 - **Multi-mode Fiber:** For shorter distances.
- **Example:** Used in backbone networks and long-distance telecommunications.
- **Performance:**
 - Extremely high bandwidth (up to Tbps).
 - Immune to electromagnetic interference.
 - Can transmit signals over very long distances with minimal loss.
- **Applications:**
 - High-speed internet, data centers, and backbone networks.
 - Used in telecommunication and military communications.

Briefly describe different types of propagation modes in optical fiber communication.

Propagation mode refers to the way signals travel through a medium, whether in wired or wireless communication. It defines how waves (electromagnetic, radio, or optical) propagate in different environments

The main types are:

1. Single-Mode Propagation

- Uses a **small core diameter** ($\approx 8\text{-}10\text{ }\mu\text{m}$).
- Allows **only one light mode** to travel.
- Has **low dispersion** and can transmit data over long distances with high bandwidth.
- Used in **long-distance communication and high-speed networks**.

2. Multi-Mode Propagation

- Uses a **larger core diameter** ($\approx 50\text{-}62.5\text{ }\mu\text{m}$).
- Allows **multiple light modes** to travel simultaneously, leading to **modal dispersion**.
- Suitable for **short-distance communication** (e.g., LANs, data centers).

Types of Multi-Mode Propagation

- **Step-Index Multi-Mode:** Light rays reflect abruptly at the core-cladding boundary, causing more dispersion.
- **Graded-Index Multi-Mode:** Core refractive index gradually decreases from center to edges, reducing dispersion and improving performance.

Each mode is used based on application requirements such as distance, speed, and cost

Define connectors. Briefly describe different types of connectors used in coaxial cable connections

Definition of Connectors:

A **connector** is a device used to join electrical or optical cables to transmit signals efficiently. In coaxial cables, connectors ensure proper signal transmission with minimal loss and interference.

Types of Connectors Used in Coaxial Cable Connections:

1. **BNC (Bayonet Neill–Concelman) Connector**
 - **Description:** A quick-connect, bayonet-style RF connector.

- **Use:** Video transmission (CCTV, broadcast), RF applications, test equipment.
- **Impedance:** 50Ω (for RF) and 75Ω (for video signals).

2. F-Type Connector

- **Description:** A threaded connector used in TV and satellite systems.
- **Use:** Cable television, satellite receivers, broadband internet.
- **Impedance:** 75Ω (optimized for video and broadband).

3. N-Type Connector

- **Description:** A durable, threaded RF connector designed for high-frequency signals.
- **Use:** Wireless communication, antennas, microwave systems.
- **Impedance:** 50Ω or 75Ω versions available.

4. TNC (Threaded Neill–Concelman) Connector

- **Description:** A threaded version of the BNC connector, offering better resistance to vibration.
- **Use:** Military, aerospace, mobile communication.
- **Impedance:** 50Ω, used in RF and microwave applications.

2. Unguided Media (Wireless Media)

Unguided media transmit electromagnetic waves without using a physical conductor. The signals are transmitted through the air or vacuum.

Types of Unguided Media:

1. Radio Waves:

- Can travel long distances and penetrate through walls.
- Used for radio and television broadcasting, mobile communications, and satellite transmissions.
- **Example:** FM radio, Wi-Fi networks.

2. Microwaves:

- Require line-of-sight communication.
- Microwaves are used for unicast communication such as cellular telephones satellite networks telephones, satellite networks, and wireless LANs.
- **Example:** Satellite networks, point-to-point communication systems.

3. Infrared (IR):

- Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation. Requires a line-of-sight and is not suitable for outdoor communication.
 - **Example:** Remote controls, short-range data transmission between devices.
-

Practical Example:

- **Home Internet Setup:**
 - **Twisted-Pair Cable** connects your modem to your router.
 - **Coaxial Cable** is used by your ISP to bring the internet connection to your home.
 - **Wi-Fi (Radio Waves)** allows your devices to connect wirelessly within your home.
- **Television Broadcasting:**
 - **Radio Waves** transmit signals from a broadcasting station to your home.
 - **Coaxial Cable** connects the antenna to the television.

Chapter-08

Why is Switching Needed?

Switching is essential in communication networks to efficiently route data from the source to the destination. It enables the sharing of network resources, ensuring that multiple devices can communicate over the same infrastructure without direct physical connections. Switching helps:

- **Optimize resource usage** by dynamically managing data paths.
- **Ensure scalability** for large networks.
- **Improve reliability** through alternative paths for data transmission.

Comparison of Switching Techniques

Feature	Circuit Switching	Packet Switching	Message Switching
Connection Setup	Required before transmission	Not required	Not required
Resource Allocation	Dedicated path	Shared path	Store-and-forward
Data Transmission	Continuous, in order	Packets may arrive out of order	Entire message is stored and then forwarded
Efficiency	Less efficient (idle time if no data)	More efficient	Can introduce delays
Example	Telephone network	Internet (TCP/IP)	Email systems

1.	Define burst error. Briefly explain a two-dimensional parity check code for error detection.	5																																								
2.	Suppose, you are provided the following Hamming code C(7,4) for error correction. <table><tr><th>Dataword</th><th>Codeword</th><th>Dataword</th><th>Codeword</th><th>Dataword</th><th>Codeword</th><th>Dataword</th><th>Codeword</th></tr><tr><td>0000</td><td>0000000</td><td>0100</td><td>0100011</td><td>1000</td><td>1000110</td><td>1100</td><td>1100101</td></tr><tr><td>0001</td><td>0001101</td><td>0101</td><td>0101110</td><td>1001</td><td>1001011</td><td>1101</td><td>1101000</td></tr><tr><td>0010</td><td>0010111</td><td>0110</td><td>0110100</td><td>1010</td><td>1010001</td><td>1110</td><td>1110010</td></tr><tr><td>0011</td><td>0011010</td><td>0111</td><td>0111001</td><td>1011</td><td>1011100</td><td>1111</td><td>1111111</td></tr></table> Explain the following cases using Hamming coding at the receiver side: - The codeword 0100011 is received. - The codeword 0011001 is received. - The codeword 0001000 is received.	Dataword	Codeword	Dataword	Codeword	Dataword	Codeword	Dataword	Codeword	0000	0000000	0100	0100011	1000	1000110	1100	1100101	0001	0001101	0101	0101110	1001	1001011	1101	1101000	0010	0010111	0110	0110100	1010	1010001	1110	1110010	0011	0011010	0111	0111001	1011	1011100	1111	1111111	9
Dataword	Codeword	Dataword	Codeword	Dataword	Codeword	Dataword	Codeword																																			
0000	0000000	0100	0100011	1000	1000110	1100	1100101																																			
0001	0001101	0101	0101110	1001	1001011	1101	1101000																																			
0010	0010111	0110	0110100	1010	1010001	1110	1110010																																			
0011	0011010	0111	0111001	1011	1011100	1111	1111111																																			
3.	Suppose CRC coding is used in a data communication system where the sender and the receiver agreed to use 1011 as the divisor. If the sender sends 1001 and the receiver receives 1000110, apply the CRC coding system for error detection in this case.	6																																								

Q1: Define burst error. Briefly explain a two-dimensional parity check code for error detection.

Burst Error:

A burst error occurs when two or more bits in a data sequence are altered due to noise or interference. Unlike a single-bit error, a burst error affects multiple bits in a sequence, typically in a contiguous manner.

Two-Dimensional Parity Check:

A two-dimensional parity check is an error detection method that organizes data into a grid format and applies parity checks to both rows and columns.

Steps:

- 1. Arrange Data in a Grid:** Place data bits in a 2D matrix.
- 2. Calculate Row Parity:** Compute parity for each row and add it as an extra bit.
- 3. Calculate Column Parity:** Compute parity for each column and add it as an extra bit.
- 4. Send the Matrix:** Transmit the entire matrix, including row and column parities.
- 5. Error Detection:** At the receiver end, recompute parity and compare with received parities to identify errors.

Solution for Hamming Code Error Detection and Correction

Understanding the Positions in Hamming (7,4) Code

Hamming (7,4) code consists of 7-bit codewords with:

- 3 Parity Bits: P_0, P_1, P_2
- 4 Data Bits: B_0, B_1, B_2, B_3

Bit Positions in Codeword:

Position	1	2	3	4	5	6	7
Bit Type	P_0	P_1	B_2	P_2	B_0	B_3	B_1

Parity Bits Cover These Positions:

- P_0 covers $\rightarrow 1, 3, 5, 7 \rightarrow (P_0, B_2, B_0, B_1)$
- P_1 covers $\rightarrow 2, 3, 6, 7 \rightarrow (P_1, B_2, B_3, B_1)$
- P_2 covers $\rightarrow 4, 5, 6, 7 \rightarrow (P_2, B_0, B_3, B_1)$

(i) Received Codeword: 0001011

Step 1: Identify the Parity Bits

From 0001011, the bits are assigned as follows:

Position	1	2	3	4	5	6	7
Bit Type	P_0	P_1	B_2	P_2	B_0	B_3	B_1
Received	0	0	0	1	0	1	1

Step 2: Compute Parity Check Syndrome

Using the parity bit rules:

- $S_0 = P_0 \oplus B_2 \oplus B_0 \oplus B_1$
 $\rightarrow S_0 = 0 \oplus 0 \oplus 0 \oplus 1 = 1$
- $S_1 = P_1 \oplus B_2 \oplus B_3 \oplus B_1$
 $\rightarrow S_1 = 0 \oplus 0 \oplus 1 \oplus 1 = 0$
- $S_2 = P_2 \oplus B_0 \oplus B_3 \oplus B_1$
 $\rightarrow S_2 = 1 \oplus 0 \oplus 1 \oplus 1 = 1$

Step 3: Determine the Error Position

The syndrome is $S_2 S_1 S_0 = 101$ (decimal 5).

So, the error is at position 5 (B_0).

Step 4: Correct the Error

Flip B_0 ($0 \rightarrow 1$), so the corrected codeword is:

0001111

(ii) Received Codeword: 0011001

Step 1: Identify the Parity Bits

From 0011001, the bits are assigned as follows:

Position	1	2	3	4	5	6	7
Bit Type	P_0	P_1	B_2	P_2	B_0	B_3	B_1
Received	0	0	1	1	0	0	1

Step 2: Compute Parity Check Syndrome

Using the parity bit rules:

- $S_0 = P_0 \oplus B_2 \oplus B_0 \oplus B_1$
 $\rightarrow S_0 = 0 \oplus 1 \oplus 0 \oplus 1 = 0$
- $S_1 = P_1 \oplus B_2 \oplus B_3 \oplus B_1$
 $\rightarrow S_1 = 0 \oplus 1 \oplus 0 \oplus 1 = 0$
- $S_2 = P_2 \oplus B_0 \oplus B_3 \oplus B_1$
 $\rightarrow S_2 = 1 \oplus 0 \oplus 0 \oplus 1 = 0$

Step 3: Determine the Error Position

The syndrome is $S_2 S_1 S_0 = 000$.

Since all bits are 0, there is no error.

So, the received codeword 0011001 is already correct.

c)

